

```

import pandas as pd
import numpy as np

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

```

```
x = data.drop('Class' , axis = 1)
```

```
print(x)
```

	Variance	Skewness	Curtosis	Entropy
0	3.62160	8.66610	-2.8073	-0.44699
1	4.54590	8.16740	-2.4586	-1.46210
2	3.86600	-2.63830	1.9242	0.10645
3	3.45660	9.52280	-4.0112	-3.59440
4	0.32924	-4.45520	4.5718	-0.98880
...	...	...	...	...
1367	0.40614	1.34920	-1.4501	-0.55949
1368	-1.38870	-4.87730	6.4774	0.34179
1369	-3.75030	-13.45860	17.5932	-2.77710
1370	-3.56370	-8.38270	12.3930	-1.28230
1371	-2.54190	-0.65804	2.6842	1.19520

```
[1372 rows x 4 columns]
```

```
y = data['Class']
```

```
print(y)
```

0	0
1	0
2	0
3	0
4	0
...	...
1367	1
1368	1
1369	1
1370	1
1371	1

```
Name: Class, Length: 1372, dtype: int64
```

```
# Test Size = 20%
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.20)
```

```
model = SVC(kernel='rbf')
```

```
model.fit(x_train, y_train)
```

```
y_pred = model.predict(x_test)
```

```
print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 1.0

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	154
1	1.00	1.00	1.00	121
accuracy			1.00	275
macro avg	1.00	1.00	1.00	275
weighted avg	1.00	1.00	1.00	275

Confusion Matrix:

```
[[154  0]
 [  0 121]]
```

Test Size = 30

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.30)
model = SVC(kernel='rbf')
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
```

```
print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

Linear Kernel SVM Accuracy: 0.9927184466019418

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.99	0.99	230
1	0.98	1.00	0.99	182
accuracy			0.99	412
macro avg	0.99	0.99	0.99	412
weighted avg	0.99	0.99	0.99	412

Confusion Matrix:

```
[[227  3]
 [  0 182]]
```

Test Size = 35

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
```

```

test_size=0.35)
model = SVC(kernel='rbf')
model.fit(x_train, y_train)
y_pred = model.predict(x_test)

print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

```

Linear Kernel SVM Accuracy: 1.0

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	250
1	1.00	1.00	1.00	231
accuracy			1.00	481
macro avg	1.00	1.00	1.00	481
weighted avg	1.00	1.00	1.00	481

Confusion Matrix:

```

[[250  0]
 [ 0 231]]

```

Test Size = 40

```

x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size=0.40)
model = SVC(kernel='rbf')
model.fit(x_train, y_train)
y_pred = model.predict(x_test)

print("Linear Kernel SVM Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test,
y_pred))
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

```

Linear Kernel SVM Accuracy: 0.9963570127504554

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.99	1.00	297
1	0.99	1.00	1.00	252
accuracy			1.00	549
macro avg	1.00	1.00	1.00	549
weighted avg	1.00	1.00	1.00	549

Confusion Matrix:

```
[[295  2]  
[  0 252]]
```